AI Based Fake News Detection Tool

AI/ML Internship Project

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Format

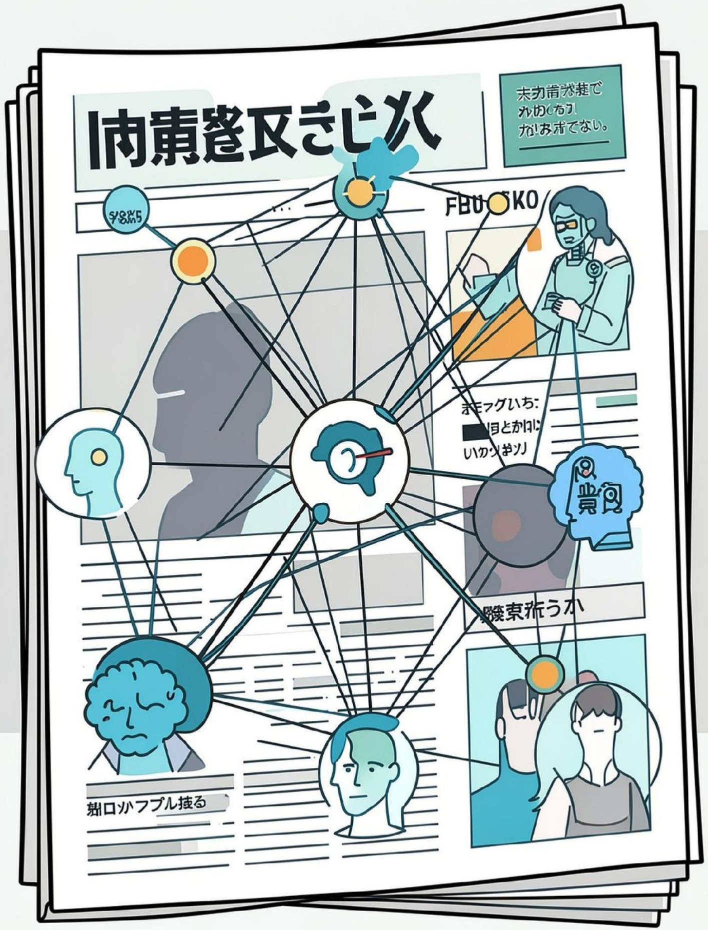
Editable slides for PowerPoint

Theme

Blue, white, dark-navy — professional

Assets

Icons, diagrams, placeholders



Project Overview

This project applies Machine Learning and Natural Language Processing to classify news articles as **REAL** or **FAKE**.

- Text features: TF-IDF vectorization
- Classifier: Logistic Regression
- Persistence: Trained model saved with Joblib

Target audience: technical evaluators — reproducible code and editable slides.

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Problem Statement

Rapid propagation

False articles spread quickly across platforms.

Manual verification slow

Human fact-checking cannot scale in real time.

Need for automation

Automated, explainable detection to assist reviewers.

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Objectives

1

Detect Fake News

Binary classification: REAL vs FAKE

2

Apply NLP

Preprocess text, TF-IDF features

3

Train Model

Logistic Regression with validation

4

Provide Confidence

Probability score for predictions

5

Save Artifact

Export model with Joblib for deployment

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Technologies Used

Python

Core language for data and model code

Pandas

Data ingestion and preprocessing

Scikit-learn

TF-IDF, Logistic Regression, metrics

Joblib

Model serialization

CSV Dataset

Text corpus labeled REAL / FAKE

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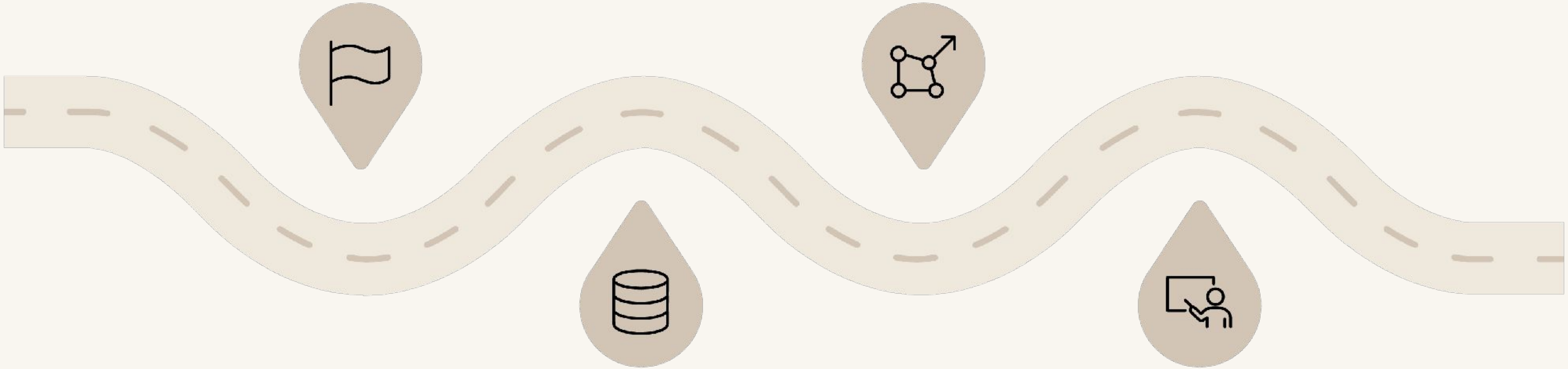
System Workflow



Start

Begin pipeline

Data Prep

Load and clean dataset

Feature

TF-IDF vectorization

Train

Train logistic regression

Caption: End-to-end pipeline from data ingestion to live prediction and confidence reporting.

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Dataset

Column	Description
text	Article content (string)
label	Target: REAL or FAKE
source	Optional: original URL or publisher

Dataset stored as CSV. Balanced sampling and train/validation split used during experiments.

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Machine Learning Model

- **TF-IDF Vectorization**

  Converts tokenized text to weighted numeric features

- **Logistic Regression**

  Interpretable linear classifier producing probabilities

- **Evaluation**

  Precision, recall, F1-score, and ROC-AUC reported

- **Output**

  Class label + confidence score (probability)

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Features & User Functions

View Dataset

Inspect raw data rows

Dataset Summary

Counts, class distribution

Train Model

Run training with logs

Detect Fake News

Single or batch prediction

Prediction Confidence

Probability shown with result

Prediction History

Audit trail for review

Exception Handling

Robust input validation and logging

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Outputs, Advantages & Future Scope

Advantages

- Fast automated detection
- Interpretable model with confidence scores
- Reusable serialized model for deployment

Future Scope

- Flask web app for online predictions
- Multi-language support & live news API
- Upgrade to Deep Learning: BERT/LSTM for improved accuracy

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